

Forest Layers and Adaptations

Tropical Rainforests and Mangrove Forests

1. Layers of the Tropical Rainforest

A – The Emergent Layer

- This is the topmost layer, and is made up of the tallest trees called emergent trees scattered throughout the forest.
- They grow up to a height of 40m – 50m to reach out for sunlight.
- They have thick and straight trunks with buttress roots to support their size.

B – The Canopy Layer

- Trees in the canopy layer grow up to a height of 20m – 30m.
- The trees in the canopy layer grow very close together and their crowns form a continuous layer called the canopy.
 - ↳ This prevents most of the sunlight from reaching the forest floor.
 - ↳ They can block up to 70% – 90% of sunlight from the forest floor.

C – The Undergrowth Layer

- Experiences very little sunlight due to the canopy layer.
- Sparse vegetation – not many plants grow here except smaller trees that are shade tolerant or seedlings of taller trees.
- Saprophytes are found here. They do not photosynthesise and live on dead and dying organic matter.
 - ↳ Examples include fungi and mushrooms which help to break down leaf litter and dead organic matter on the forest floor.

2. Adaptations of Tropical Rainforest Plants

Leaves

- Leaves at the bottom layers are generally broad to take in as much sunlight as possible.
- They are waxy and smooth to repel rainwater.
- Many have drip tips to allow water to drip off easily.

These adaptations prevent water from collecting on the leaves which may encourage bacterial growth.

Bark

Bark is generally smooth and thin as there is no necessity for protection against extreme cold or dry weather.

Roots

- The roots are shallow as the high rainfall provides most of the water, which can be found in the top layer of the soil. Nutrients in the tropical forests are also found in the top layer of the soil because of the rapid decay of leaves, twigs and branches.
- ↳ This results in poor stability of the trees.
- The larger trees thus develop buttress roots as well to provide additional support to the trees.
- ↳ These structures can reach 1m – 5m above the ground.

Branches

- Most trees have wide-spread branches like umbrellas as they spread out their leaves to obtain the most amount of sunlight.
- These are typically found at the top one-third of the trees so that they are able to obtain maximum light for photosynthesis.

Flowers and Fruits

- Fruits and flowers are normally brightly coloured and strong-smelling to attract animals.
- ↳ As the air is generally still in the dense forest, the vegetation mainly depends on animal dispersal (animals to pollinate their flowers or spread their seeds).

3. Mangroves

Mangrove forests have dense vegetation because of high temperatures and rainfall. Similar to tropical rainforests, the canopy is dense and little sunlight gets through to reach the ground, resulting in sparse undergrowth. Trees vary from 2–40m as the muddy soil cannot provide firm support.

There is a lower diversity of plants in the mangrove forest as compared to tropical rainforests. Only a small number of trees can survive in the salty conditions along the coast. Mangrove trees, however, are halophytes, which are highly specialised plants that have adapted to growing in salty conditions.

Tropical mangroves provide valuable resources such as food, medicine, firewood, construction materials and charcoal.

Structure – Horizontal Zones

Mangrove forests are divided into three distinct horizontal zones:

- 1 Coastal Zone
- 2 Middle Zone
- 3 Inland Zone

The following factors increase towards the sea: salinity, inundation (amount of time it is submerged under water), soil instability, and sedimentation rate.

Zone	Mangrove Species Found
Coastal Zone	Avicennia, Sonneratia
Inland Zone	Rhizophora, Bruguiera

(As listed in the source material, species are grouped by Coastal and Inland zones.)

Adaptations

- Mangrove plants have aerial roots that help them to take in oxygen directly from the air when they are exposed, and anchor them to the soft soil so that they will not be washed away.
 - In order to survive in a highly saline environment, mangrove plants have also developed salt-secreting leaves and salt-excluding roots.
 - Fruits and flowers are adapted for water and animal dispersal.
 - Mangrove tree flowers are generally colourful in order to attract animals/insects.
 - Because mangrove trees live near the sea, some of their fruits or seeds are buoyant and can float on water. This allows them to be carried away on the water to other areas of the coast where they can germinate.
 - Due to the poor soil stability and fast-moving waters in the coastal zone, mangroves can be viviparous.
- ↳ Some plants begin to germinate and grow while still on the parent tree, so that the propagules can be dispersed via water to increase their chances of survival.